Beyond PII Removal: Risk-Budgeted Generalization for LLM-Ready Sensitive Text

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Paper under double-blind review

Abstract

Sensitive text corpora are valuable for training, retrieval, evaluation, and domain adaptation, but common de-identification pipelines often focus on direct PII spans such as names, emails, and phone numbers. We study residual inference risk from quasi-identifiers: dates, locations, occupations, schools, demographic attributes, medical/legal/financial facts, family relations, and combinations that remain informative after direct redaction. We propose Risk-Budgeted Quasi-Identifier Generalization (RB-QIG), a lightweight transformation layer that scores quasi-identifiers by privacy risk and utility importance, then applies the least destructive generalization needed to satisfy a document-level risk budget. On a controlled synthetic benchmark, RB-QIG balanced reduces risk-weighted leakage to 23.6% [22.9, 24.5] while preserving 71.7% [70.8, 72.4] of utility facts; blanket quasi-identifier redaction removes leakage but preserves only 43.3% [39.9, 46.8] of utility facts. On a 100-record English RAT-Bench pilot, direct redaction leaves 78.0% [73.4, 82.5] LLM-attacker risk-weighted leakage and a naive LLM sanitizer leaves 46.0% [40.1, 52.0], while RB-QIG balanced reduces it to 5.7% [3.2, 8.8]. The result is not an anonymization guarantee; it is evidence that responsible data transformation should evaluate what capable models can still infer, not only whether obvious PII spans were removed.

1 Introduction

Sensitive text can improve foundation-model systems, but direct use is often blocked by privacy risk. Practical de-identification systems tend to prioritize direct identifiers: names, emails, phone numbers, account numbers, and similar spans. This is necessary, but incomplete. A record can contain no explicit name and still reveal a person or sensitive attribute through combinations of age, date, location, occupation, institution, health condition, legal event, family role, or demographic status. Recent work on LLM inference attacks shows that models can infer personal attributes from ordinary text even when the attribute is not explicitly stated Staab et al. (2024). RAT-Bench similarly argues that text anonymization should be evaluated by residual re-identification risk, not only span recall Krčo et al. (2026).

We study a simple question: after direct PII is removed, can an auditable generalization layer reduce residual quasi-identifier risk while retaining more utility than blanket redaction? We introduce RB-QIG, a risk-budgeted quasi-identifier generalization policy. RB-QIG extracts candidate quasi-identifiers, assigns each a privacy risk and utility importance, and then chooses generalizations under a document-level risk budget. The goal is not to certify anonymity. The goal is a practical transformation and evaluation protocol for sensitive text that may later be used in retrieval, training, evaluation, or auditing pipelines.

Our contributions are:

- a quasi-identifier taxonomy for LLM-ready sensitive text;
- an auditable risk-budgeted generalization algorithm;
- a low-cost evaluation protocol using deterministic leakage metrics, bootstrap intervals, and cached LLM attackers; and

- a privacy-utility comparison against no redaction, direct redaction, naive LLM sanitization, and blanket quasi-identifier redaction.

2 Threat Model and Method

We assume an attacker receives transformed text, can use a capable LLM, and tries to infer benchmark attributes such as date of birth, demographic status, education, location, employment status, family structure, or sensitive domain facts. The attacker is restricted to benchmark attributes and is not asked to identify real people. All experiments use synthetic or public benchmark data.

Definitions. A direct identifier is a span that directly identifies a person or account, such as a name, email, phone number, address, or account number. A quasi-identifier is a span that may not identify a person alone but can identify or reveal private attributes in combination with other spans. Examples include exact dates, fine-grained locations, rare occupations, employers, schools, diagnoses, legal events, financial events, family relationships, and rare narrative facts. A generalization replaces a precise value with a broader but still useful category, such as "Boise, Idaho" to "a city in the U.S. Mountain West" or "oncology infusion pharmacist" to "clinical pharmacy professional."

RB-QIG. Each quasi-identifier q_i receives privacy risk $r_i \in [0, 5]$ and utility importance $u_i \in [0, 5]$. Each candidate has a short transformation ladder: keep, generalize, or redact. RB-QIG uses a document risk score

$$R(x) = \sum_i r_i + 0.5 \binom{|\{i : r_i \geq 3\}|}{2},$$

where the second term approximates combination risk from multiple linkable attributes. It greedily applies the next transformation with largest risk drop per utility loss until the risk budget is met or no transformations remain. All edits are span replacements and every replacement is logged.

Baselines. We compare no transformation, direct PII redaction, a naive LLM sanitizer, blanket quasi-identifier redaction, and RB-QIG with strict, balanced, and utility-oriented budgets. Direct redaction uses regexes plus benchmark-provided direct identifiers. The LLM sanitizer receives the direct-redacted text and a simple instruction to remove personally identifying information while preserving meaning. Presidio, for example, is a practical de-identification SDK, but its documentation warns that automated detection cannot guarantee finding all sensitive information Microsoft (2026).

3 Experiments

Synthetic control. We generate 100 synthetic records across medical, legal, financial, workplace, and education-support domains. Each record has direct identifiers, quasi-identifiers, ground-truth attributes, a utility label, and utility facts. Because oracle quasi-identifier spans are known, this setting isolates the transformation tradeoff.

RAT-Bench pilot. We evaluate 100 English difficulty-1 RAT-Bench records Krčo et al. (2026). We use a cached target-aware extractor to find value-revealing spans and variants for benchmark target attributes. This is a controlled transformation benchmark: given target quasi-identifiers, how should they be transformed? It is not a claim that the extractor is deployment-ready. We also run a blind extractor over the same 100 records and score against benchmark QIs as a coverage stress test, with a generic backstop for common demographic, citizenship, education, employment, marital, and race/ethnicity cues.

LLM attacker and utility judge. For RAT-Bench, a cached LLM attacker receives transformed text and requested attribute names, then returns structured guesses with evidence.

For utility, cached LLM judges compare original and transformed text and score label preservation, fact preservation, and semantic utility without rewarding preserved direct identifiers. All API calls use public or synthetic benchmark records only, are cached, and cost approximately $1.85 total across the current workspace artifacts.

4 Results

Table 1 shows the controlled privacy-utility frontier. Direct redaction preserves utility but leaves all quasi-identifiers. Blanket quasi-identifier redaction eliminates leakage but preserves only 43.3% of utility facts. RB-QIG balanced preserves 71.7% of utility facts while reducing risk-weighted leakage from 100.0% to 23.6%. The paired bootstrap advantage in utility-fact preservation over blanket redaction is +28.3 points [25.2, 31.6].

Table 3 gives a more forgiving semantic utility read. The LLM judge treats placeholders and broad context as partially useful; under this metric, RB-QIG balanced has a modest semantic-utility edge over blanket redaction: +2.6 points [0.6, 4.8]. On blind-backstop RAT-Bench, however, the same judge finds no meaningful RB-QIG edge over blanket redaction: -0.4 semantic-utility points [-4.2, 3.4]. We therefore make the large utility claim from the stricter controlled utility-fact metric and treat public utility as a caveat. A no-API annotation-specificity diagnostic still shows RB-QIG keeps more typed or generalized QI semantics than blanket placeholders: +6.8 points [4.7, 9.2] on blind RAT-Bench and +6.2 [5.5, 6.9] on TAB, but this is not task accuracy.

Table 2 confirms the residual-risk problem on public data. After direct redaction, the LLM attacker still recovers many target attributes: 78.0% risk-weighted leakage. A naive LLM sanitizer improves over direct redaction but still leaves 46.0% leakage. RB-QIG balanced reduces leakage to 5.7%, a paired drop of 72.2 points [67.0, 77.4] relative to direct redaction and 40.2 points [33.1, 47.1] relative to the LLM sanitizer. RB-QIG and blanket redaction are statistically tied on this privacy metric: +0.5 points [-2.6, 3.6]. The current public pilot therefore supports the reduction relative to direct or naive LLM redaction rather than a privacy advantage over blanket redaction.

A model-free Presidio-style pattern baseline gives a practical direct-PII check. On RAT-Bench it leaves 37.0% [28.0, 46.0] direct-identifier leakage and 85.0% [80.6, 89.0] deterministic risk-weighted leakage; on TAB it leaves 100.0% direct-identifier leakage because legal names and case identifiers are not common PII patterns. This makes the oracle-assisted direct baseline conservative rather than weak.

As a harder public-data smoke, we also ran 30 English RAT-Bench difficulty-2 records. Direct and naive LLM redaction left 70.8% and 56.9% LLM-attacker risk-weighted leakage, respectively, while RB-QIG balanced reduced leakage to 13.1% [3.3, 25.0] and was statistically tied with blanket redaction. On a 20-record stronger-attacker smoke, RB-QIG balanced remained far below direct redaction (18.0% versus 77.3% leakage) and statistically tied with blanket redaction.

A no-API 30-document TAB ECHR legal-domain screen provides deterministic cross-domain evidence only: direct redaction leaves 99.8% [99.4, 100.0] risk-weighted leakage, while RB-QIG balanced lowers it to 12.3% [9.2, 15.7]. Its broad deterministic utility proxy is uninformative; a 10-document legal-task LLM utility screen is also inconclusive, with both RB-QIG and blanket redaction at 68.0% [62.0, 74.0] and paired 0.0 points [-10.0, 10.0]. We therefore use TAB as residual-risk support rather than a utility win.

Table 4 summarizes the deployment-style blind stress test. The raw blind extractor covers 72.8% of benchmark QI spans; a generic deterministic backstop raises coverage to 99.6% without additional API calls. Backstopped blind RB-QIG balanced reduces deterministic risk-weighted leakage relative to direct redaction from 94.3% [91.3, 96.8] to 5.4% [2.8, 8.5], a paired drop of 88.9 points [85.0, 92.5]. However, it is 2.3 points [0.6, 4.9] leakier than blanket quasi-identifier redaction under deterministic coarse scoring. Under the LLM attacker, the same outputs reduce risk-weighted leakage from 78.1% [73.3, 82.9] to 6.4% [3.9, 9.1], and are statistically tied with blanket redaction: -0.5 points [-2.5, 1.5]. Public semantic utility is

Table 1: Synthetic controlled benchmark. Brackets show bootstrap 95% confidence intervals.

Method	Risk-weighted leak	Utility facts	Token change
None	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
Direct	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	13.5 [12.6, 14.3]
Blanket QI	0.0 [0.0, 0.0]	43.3 [39.9, 46.8]	63.7 [62.9, 64.4]
RB-QIG strict	16.1 [15.7, 16.4]	48.3 [45.3, 51.4]	59.3 [58.7, 60.0]
RB-QIG balanced	23.6 [22.9, 24.5]	71.7 [70.8, 72.4]	59.0 [58.3, 59.7]
RB-QIG utility	38.0 [37.5, 38.4]	95.0 [93.0, 96.8]	56.6 [55.8, 57.5]

Table 2: RAT-Bench LLM attacker results. Brackets show bootstrap 95% confidence intervals.

Method	Record compromise	Exact attr leak	Coarse attr leak	Risk-weighted leak
Direct	76.0 [67.0, 84.0]	74.9 [69.5, 80.0]	81.2 [76.8, 85.4]	78.0 [73.4, 82.5]
LLM direct	34.0 [25.0, 43.0]	48.7 [42.6, 54.6]	52.8 [46.5, 58.7]	46.0 [40.1, 52.0]
Blanket QI	3.0 [0.0, 7.0]	3.9 [1.5, 7.0]	7.3 [4.3, 10.8]	5.3 [2.7, 8.3]
RB-QIG balanced	2.0 [0.0, 5.0]	4.0 [1.6, 6.9]	8.8 [5.3, 12.8]	5.7 [3.2, 8.8]

also tied with blanket redaction, so blind extraction and utility-preserving rewriting remain open problems rather than solved deployment results.

Failure taxonomy. Residual RB-QIG leaks are mostly relational or lexical variants: widowhood from bereavement cues, sex from gendered language, education from school-name mentions, citizenship from non-citizen phrasing, and employment or armed-forces status from occupation context. These failures are not just missed exact spans; they are cues that a model can combine. This supports the broader argument that data transformation should evaluate inferability, not only span categories.

5 Limitations and Ethics

RB-QIG is not an anonymization guarantee and should not be used as a legal or compliance claim. The main public benchmark extraction is target-aware, so it evaluates transformation behavior under known benchmark attributes rather than fully blind deployment. Blind public and synthetic diagnostics suggest that utility-aware conversion and deterministic backstops help but still leave an extraction tradeoff: on synthetic records, RB-QIG balanced preserves 43.3% [35.0, 51.7] of utility facts while leaving 5.2% [1.7, 9.7] oracle-measured risk-weighted leakage. LLM attackers and judges are low-cost automated evaluators and may under- or over-estimate real adversaries or utility needs. All experiments use synthetic or public benchmark data. We do not attempt to identify real people or infer private attributes about individuals.

6 Conclusion

Direct PII removal is an incomplete transformation for LLM-ready sensitive text. RB-QIG demonstrates that a small, auditable risk-budgeted generalization layer can dramatically reduce measured residual inference risk while preserving useful semantic content better than blanket redaction in controlled settings. The main open problem is robust blind quasi-identifier extraction that preserves task-relevant utility while handling relational and lexical inference cues.

A Reproducibility Notes

The artifact package is generated from local scripts under `src/` and public/synthetic data only. The main synthetic control is `results/synthetic_100/`; the public RAT-Bench pilot is `results/ratbench_d1_api_100/`; and the blind public stress test is `results/`

Table 3: Synthetic LLM utility judge results. Brackets show bootstrap 95% confidence intervals.

Method	Label preserved	LLM fact preservation	Semantic utility
Direct	100.0	75.3 [73.3, 77.3]	83.2 [81.8, 84.8]
Blanket QI	99.0	69.1 [66.6, 71.4]	76.8 [74.8, 78.4]
RB-QIG balanced	100.0	71.3 [69.6, 73.0]	79.4 [78.2, 80.6]

Table 4: Blind public RAT-Bench stress test. Brackets show bootstrap 95% confidence intervals.

Method	LLM risk-weighted leak	Semantic utility	LLM fact preservation
Direct	78.1 [73.3, 82.9]	86.2 [83.2, 88.8]	84.3 [81.2, 87.0]
Blanket QI	6.8 [4.0, 9.9]	62.6 [59.2, 66.0]	53.8 [49.5, 58.1]
RB-QIG balanced	6.4 [3.9, 9.1]	62.2 [58.8, 65.2]	55.5 [52.0, 58.7]

`ratbench_d1_blind_backstop_v2_budgetfix_api_100/`. Bootstrap intervals are generated by `src/bootstrap_results.py`, and paper tables by `src/make_paper_tables.py`. The qualitative appendix is generated by `src/make_qualitative_appendix.py`; headline claim checks are generated by `src/make_claim_audit.py`. Cached API usage in the workspace is approximately $1.85, and all API calls use public or synthetic benchmark records.

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